1) If the concentration of a solution is *tripled* and the length of the sample cell is *halved*, absorbance value at peak maximum will ____

(a) remain the same as the original value
(b) be ⅔ of the original value
(c) be 1.5 times the original value
(d) be six times the original value

Ans:

3) In general, the Boltzmann equation, suggests that at _____ temperatures and with _____ energy separation (ΔE), a negligibly small fraction of molecules is likely to occupy the excited state as compared to the ground state.

(a) low; small
(b) high; large
(c) high; small
(d) low; large

Ans: D

4) The position of a peak in a spectrum is determined by:

(a) Only the energy gap ΔE
(b) the energy gap ΔE and the path length
(c) the energy gap ΔE and the concentration
(d) the energy gap ΔE and the probability

Ans:

5) *Cyclopentanone* (**A**) shows two peaks in the UV-visible spectrum at 200 nm and 295 nm.
Which of the below options is correct with regard to the statements (i) - (iv)?



(A)

(i) peak at 200 nm is assigned to $n \rightarrow \pi^*$ transition
(ii) peak at 295 nm is assigned to $\pi \rightarrow \pi^*$ transition
(iii) peak at 200 nm will be stronger in intensity
(iv) peak at 295 nm will be weaker in intensity

(a) only (iii) is true
(b) (i), (ii) and (iv) are true
(c) (i) and (ii) true
(d) (iii) and (iv) are true

Ans:

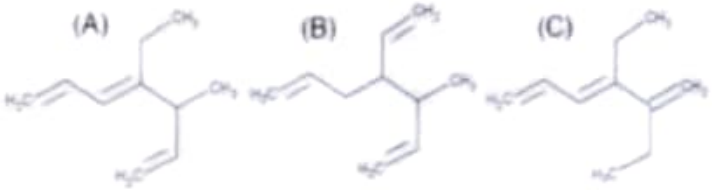
D

7) Two organic compounds, trimethylamine $(CH_3)_3N$ and trimethylammonium chloride $(CH_3)_3NH^+Cl^-$), can be distinguished from one another using UV-visible spectrum using which of the following facts?

(a) Disappearance of $n \rightarrow \pi^*$ transition
(b) Appearance of $\sigma \rightarrow \sigma^*$ transition
(c) Appearance of $n \rightarrow \pi^*$ transition
(d) Disappearance of $n \rightarrow \sigma^*$ transition

Ans:

8) The peak position (λ_{max}) of the $\pi \rightarrow \pi^*$ transition in the UV-visible absorption spectrum of the following compounds will follow the order:



(a) C > B > A
(b) B > A > C
(c) C > A > B
(d) A > B > C

Ans: